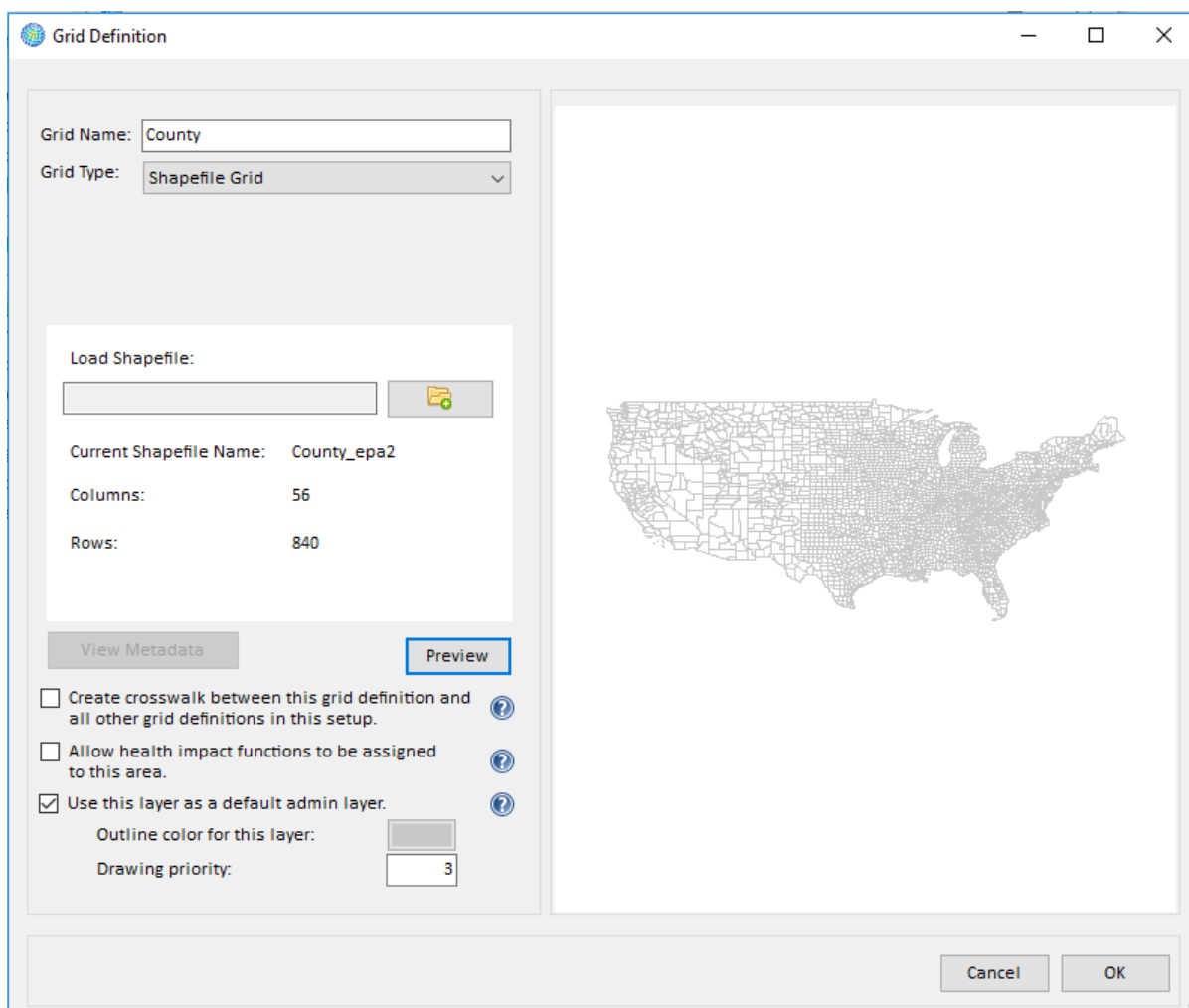




Starting with BenMAP-CE version 1.4, you can apply different health impact functions to different air quality grids within the same BenMAP run. This feature is designed to facilitate the application of region-specific air pollution effect coefficients (e.g., you can apply a Los Angeles-specific health impact function to that city and a national function to the remainder of the United States). If you plan to link specific health impact functions to a new grid definition, you must check the “Allow health impact functions to be assigned to this area” box in the lower left of the screen. Checking this box does not require you to assign a health impact function to this area, but enables this feature as an option.

Also added in version 1.4, you can now specify whether you want to use a grid definition as a default administrative (admin) layer; these layers are useful for showing political boundaries in GIS maps of BenMAP results. By checking the box “Use this layer as a default admin layer” you are telling BenMAP to draw this layer on the GIS results map by default. You may select multiple admin layers, such as a country boundary and state or province boundaries, and you can turn individual administrative layers on or off manually at any point using the GIS tool.

4.1.1.1 Regular Grid

Regular Grid Definitions are defined by a lower left corner (specified as decimal degree latitude and longitude, with West and South having negative values and East and North having positive values), a total number of columns and rows, a number of columns per degree longitude, and a number of rows per degree latitude. Individual cells within the resultant grid are numbered in sequential order (columns from left to right, rows from bottom to top) starting at (1, 1). These field values will be used to link the **Regular Grid Definition** with other sources of data, as discussed in more detail below.

To define a **Regular Grid**, start by selecting *Regular Grid* from the **Grid Type** drop-down menu. Type the name of the grid definition in the **Grid ID** box, and then define the number of **Columns** and **Rows** in the grid. To locate this grid geographically, provide the decimal degree coordinates for the lower left-hand corner of the grid in the **Minimum Longitude** and **Minimum Latitude** boxes.

To give the overall geographic size of the grid, provide the number of **Columns Per Longitude** and **Rows Per Latitude**. For example, if you specify 16 columns and 2 columns per degree longitude, then the grid will span 8 degrees of longitude. And if you specify 25 rows and 4 rows per degree latitude, then the grid will span 6.25 degrees of latitude.

Combining the numbers in this example, if the minimum longitude and latitude are -81 and 38 and the grid spans 8 degrees longitude and 6 degrees latitude, then the grid will run between -81 and -73 degrees longitude and between 38 and 44.25 degrees latitude.

After defining the grid, click the **Preview** button to see what the grid looks like. You may change the parameters and click the **Preview** again to see how the grid changes. When you are satisfied with the grid definition, click the **OK** button.

Grid Definition

Grid Name:

Grid Type:

Columns: Rows:

Minimum Longitude: Minimum Latitude:

Columns Per Longitude: Rows Per Latitude:

☐ Create crosswalk between this grid definition and all other grid definitions in this setup. ?

☐ Allow health impact functions to be assigned to this area. ?

☐ Use this layer as a default admin layer. ?

Outline color for this layer:

Drawing priority:

Fundamental Concept - Crosswalks

BenMAP-CE calculates **crosswalks** between different grid definitions to relate data at one spatial scale to another. For the default grid definitions in the U.S. BenMAP Setup, EPA developed population-weighted crosswalk factors between geographic elements using census tract-level population data. In all other cases, BenMAP-CE calculates a crosswalk allocation factor based on the percent overlap in area between a geographic element of one grid definition and every element of another grid. This allows BenMAP-CE, for example, to use population at the county level with air quality at the 12km CMAQ grid level in a health impact analysis. Crosswalks also allow BenMAP-CE to aggregate or disaggregate results to various geographic resolutions.

To calculate health impacts and economic benefits, BenMAP-CE uses air quality, population, and demographic data at different spatial scales. To do this, the program calculates a crosswalk (percentage file) that relates data at one spatial scale to another (e.g., 12km CMAQ grid to county). This step is performed only once per crosswalk and the results are saved to the database for subsequent calculations.

If you wish to pre-calculate the crosswalk (percentage overlap) between this grid definition and all other grid definitions in this setup, check the option box. This will take longer to load the data now, but can save you time later when you are ready to calculate health impacts. If you do not create the crosswalks at the grid definition stage, BenMAP-CE

will create crosswalks as needed during the configuration or aggregation, pooling, and valuing stages.

The name of your newly defined grid will then appear in the **Manage Grid Definitions** window. You may click **Edit** to change the grid definition, **Delete** to permanently remove the grid that you just defined, or **Add** to define a new grid definition. **View Metadata** is not applicable for **Regular Grid Definitions**. Click **OK** to return to the **Modify Datasets** window.

4.1.1.2 Shapefile Grid

Shapefiles used to create **Shapefile Grid Definitions** should be of the ESRI Shapefile format. Details on this format can be found at <https://www.esri.com/library/whitepapers/pdfs/shapefile.pdf>. When a shapefile is being used to create a new grid, BenMAP-CE will: (1) check to see if the file is projected to GCS NAD 83; and (2) if necessary, re-project the file (with notification) from the native projection to GCS NAD 83. Any shapefiles used must contain integer fields named Column (or Col) and Row, and each shape within the shapefile must contain a unique combination of values for these two fields. These column and row values are used, just as the Column and Row field values in Regular Grid Definitions, to link the Shapefile Grid Definition with other sources of data, as discussed in more detail below.

To add a **Shapefile Grid**, click on the **Add** button in the **Manage Grid Definitions** window, choose *Shapefile Grid* from the **Grid Type** dropdown menu, name the grid in the **Grid ID**, and browse for the correct shapefile by clicking on the small open-file icon just to the right of the **Load Shapefile** input box. After locating the file, click **Open**. This will choose the file, and bring you back to the **Grid Definition** window. To view the shapefile, click **Preview**. You can add metadata using the **View Metadata** button. Adding metadata allows you to supply a file **Reference** (e.g., person, organization, publication, or model that produced or supplied the values in the electronic file) and **Description** (or any relevant notes about the use or limitations of the data). Other minimal file attributes are pre-populated automatically during import.